REV	Description	Date
V1.0	First Release	2019-6-21

AC7Z010/AC7Z020 Schematics

ZYNQ硬 件 册

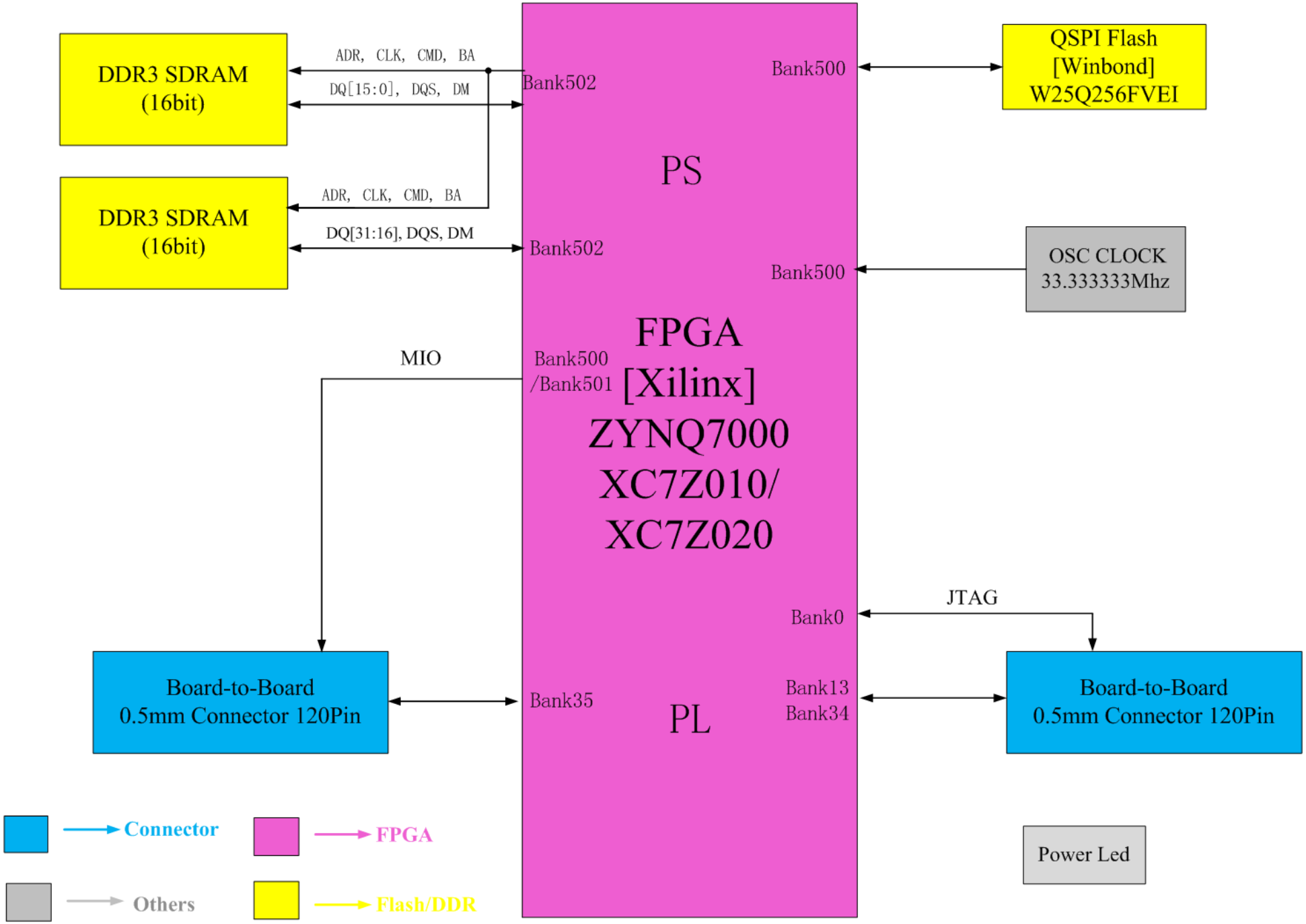
Page Number	Description
Page01	Cover Page
Page02	Block Diagram
Page03	Zynq-7000 JTAG & Bank0
Page04	Zynq-7000 MIO Config
Page05	Zynq-7000 Bank13-34-35
Page06	Zynq-7000 Bank502
Page07	Zynq-7000 Power
Page08	DDR3
Page09	GPHY
Page10	USB OTG
Page11	UART, SD
Page12	EXTEND IO
Page13	POWER

ALINX

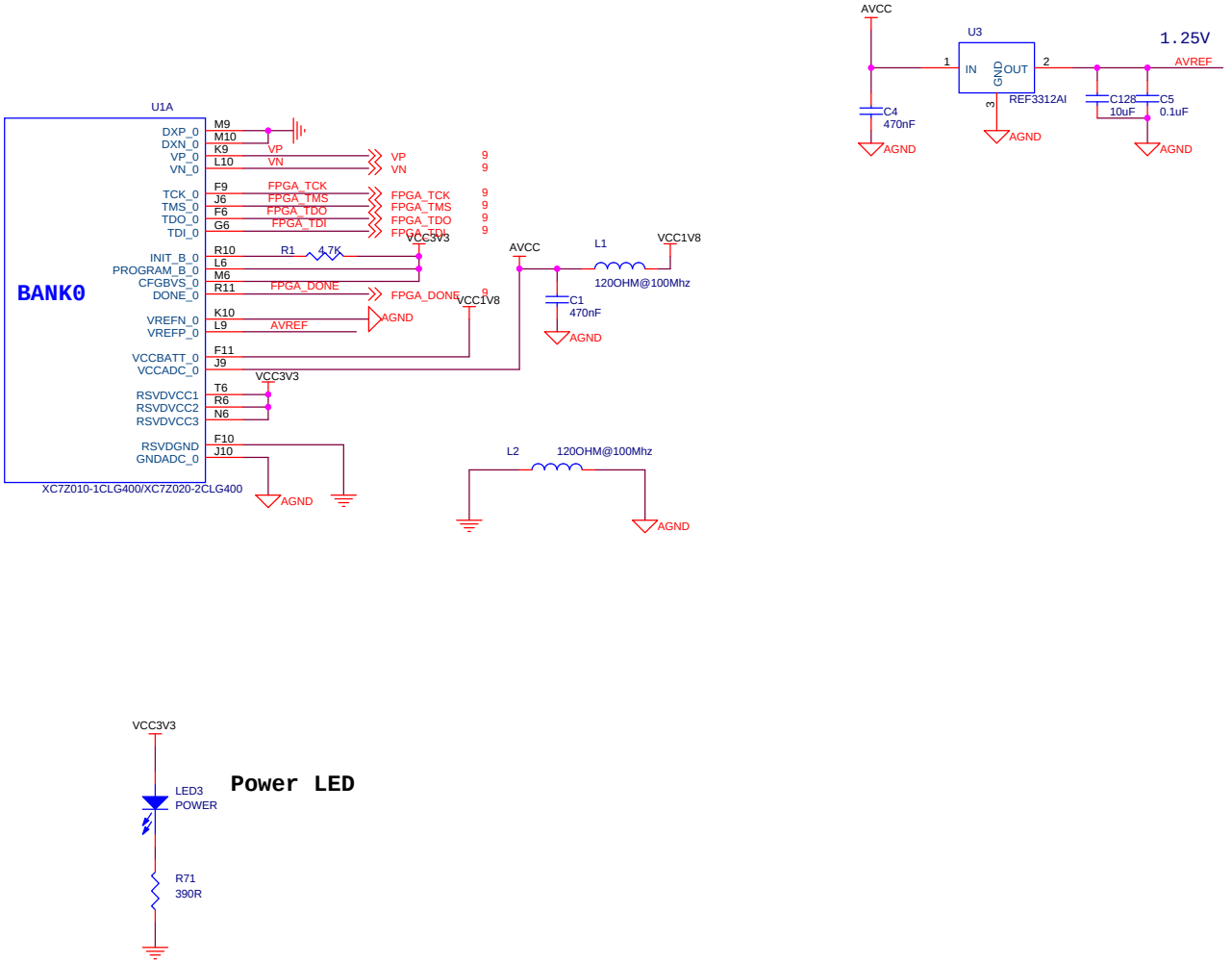
Title: Zynq-7000 Cover Page

Size | Document Number: AC7Z010/AC7Z020 核心板原理图 | Rev: 1.0

Date: Wednesday, November 30, 2022 | Sheet 1 of 13

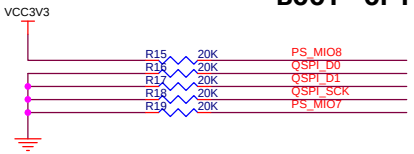


ALINX		
Title	Zynq-7000 Block diagram	
Size	Document Number: AC7Z010/AC7Z020 核心板原理图	Rev: 1.0
Date: Wednesday, November 30, 2022	Sheet 2 of 10	



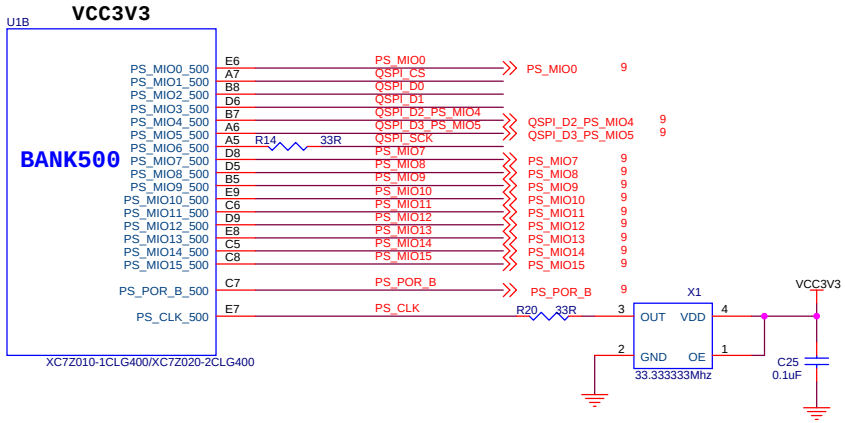
Title	Zynq-7000 JTAG & Bank0	
Size	Document Number: AC7Z010/AC7Z020 核心板原理图	Rev: 1.0
Date: Wednesday, November 30, 2022	Sheet 3 of 10	

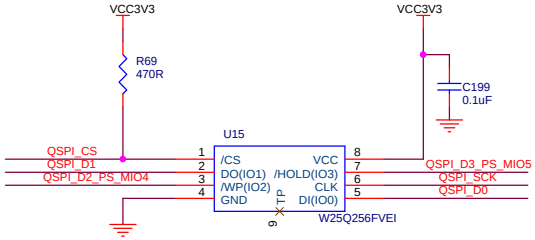
BOOT OPTION



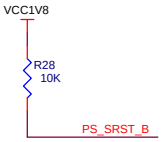

```
MIO[8] =1 ----MIO bank501 voltage=1.8V
MIO[2] =0 ----cascaded JTAG
MIO[3] =0 ----JTAG/NAND/Quad-SPI/SD
MIO[6] =0 ----PLL used
MIO[7] =0 ----MIO bank0 voltage=3.3V
```

VCC3V3

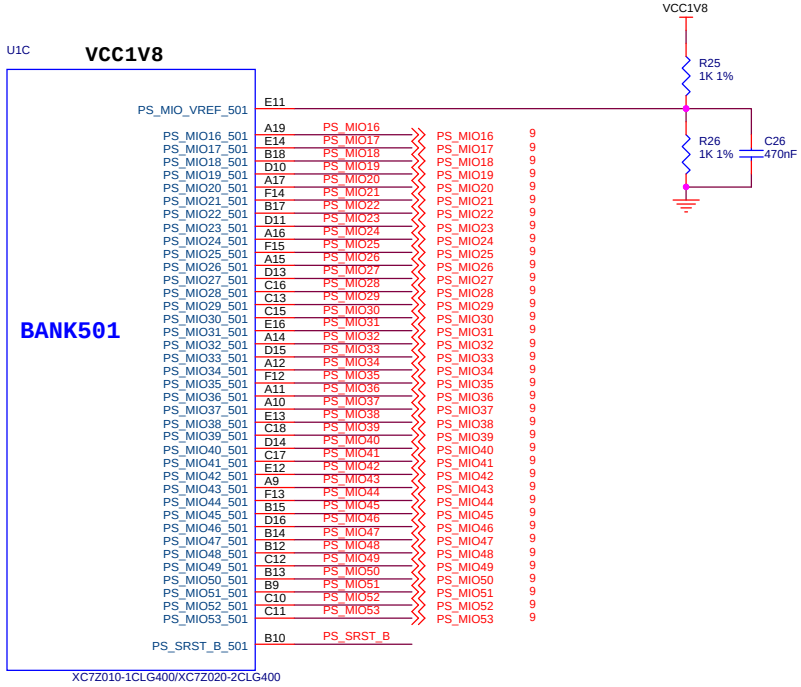




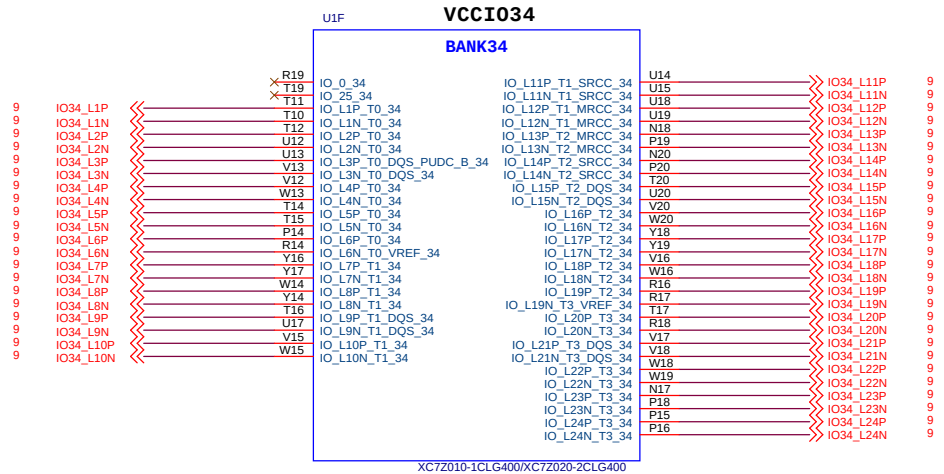
QSPI FLASH

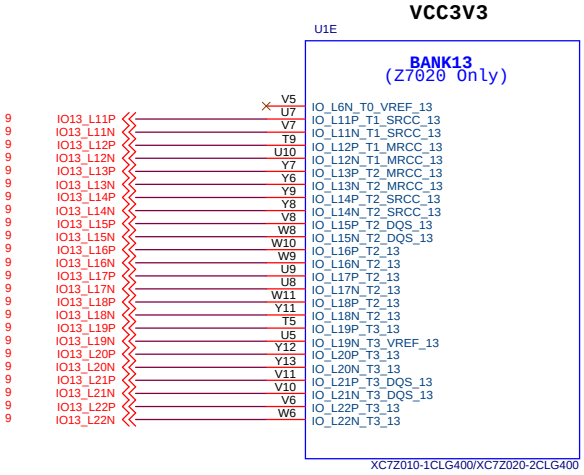


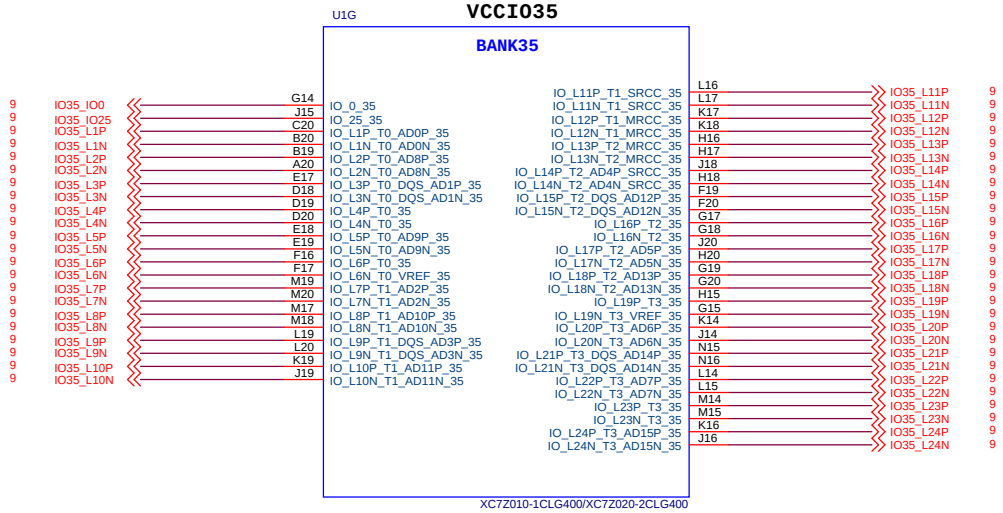
VCC1V8



Title	Zynq-7000 PS MIO	
Size	Document Number: AC7Z010/AC7Z020 核心板原理图	Rev: 1.0
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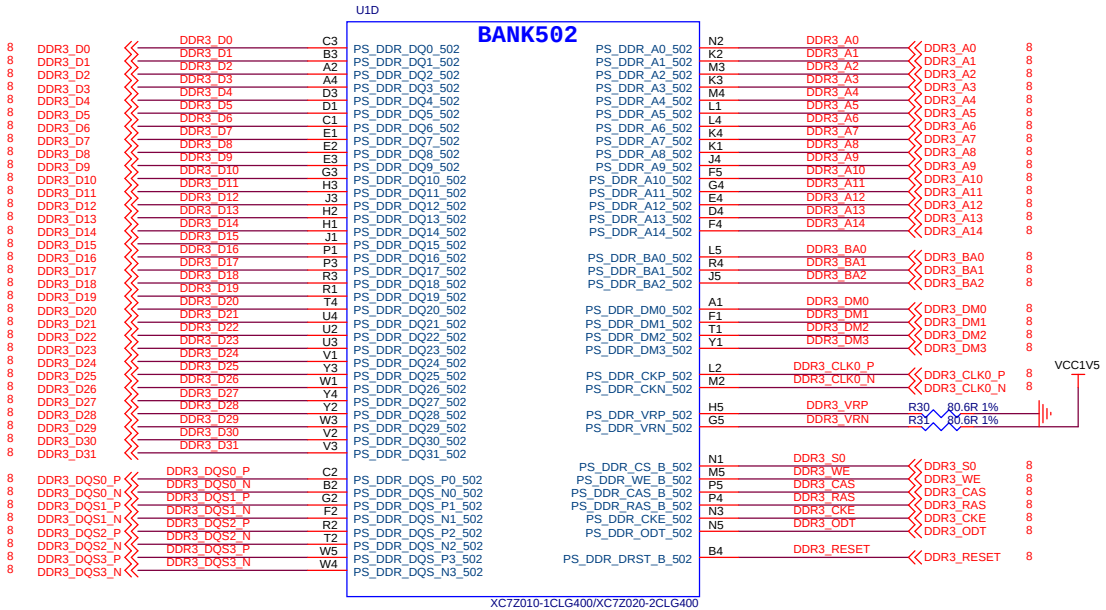


ALINX

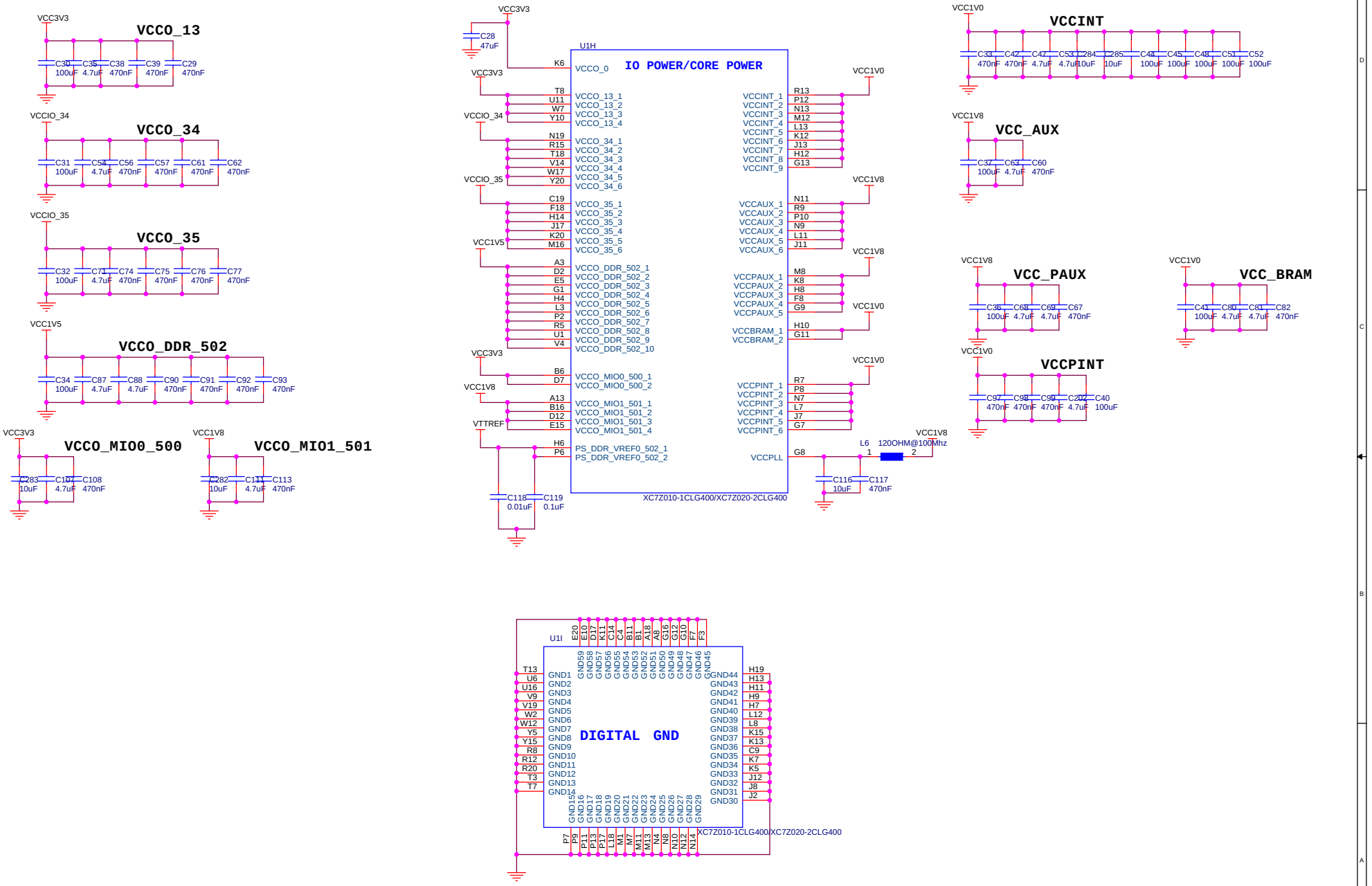
Title: Zynq-7000 Bank13 & 34 & 35

Size | Document Number: AC7Z010/AC7Z020 核心板原理图 | Rev: 1.0

Date: Wednesday, November 30, 2022 | Sheet 5 of 10

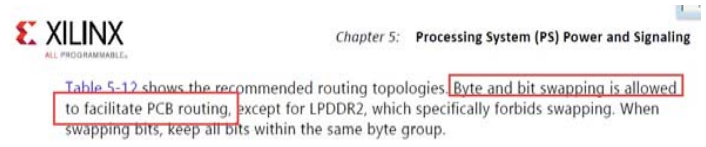


Title	Zynq-7000 Bank502
Size	
Document Number	AC7Z010/AC7Z020 核心板原理图
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Date:	Wednesday, November 30, 2022
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Title	Zynq-7000 Cover Page	
Size	Document Number: AC7Z010/AC7Z020 核心板原理图	Rev: 1.0
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为了 PCB走线方便，DDR3数据线组内（DQS除外）可以任意交换。
XILINX 的UG933文档的PAGE66页以下说明



Chapter 5: Processing System (PS) Power and Signaling

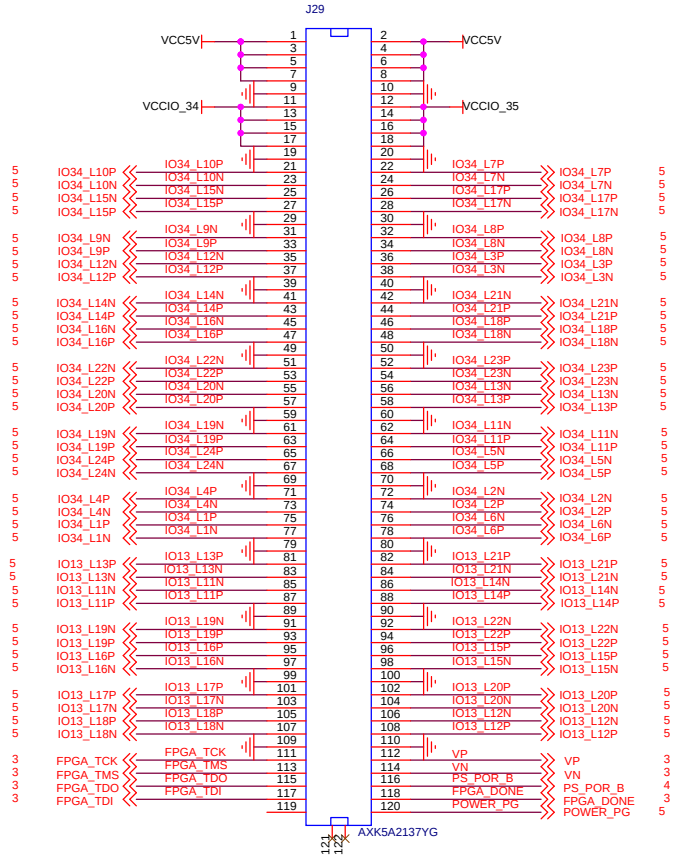
Table 5-12 shows the recommended routing topologies. Byte and bit swapping is allowed to facilitate PCB routing, except for LPDDR2, which specifically forbids swapping. When swapping bits, keep all bits within the same byte group.

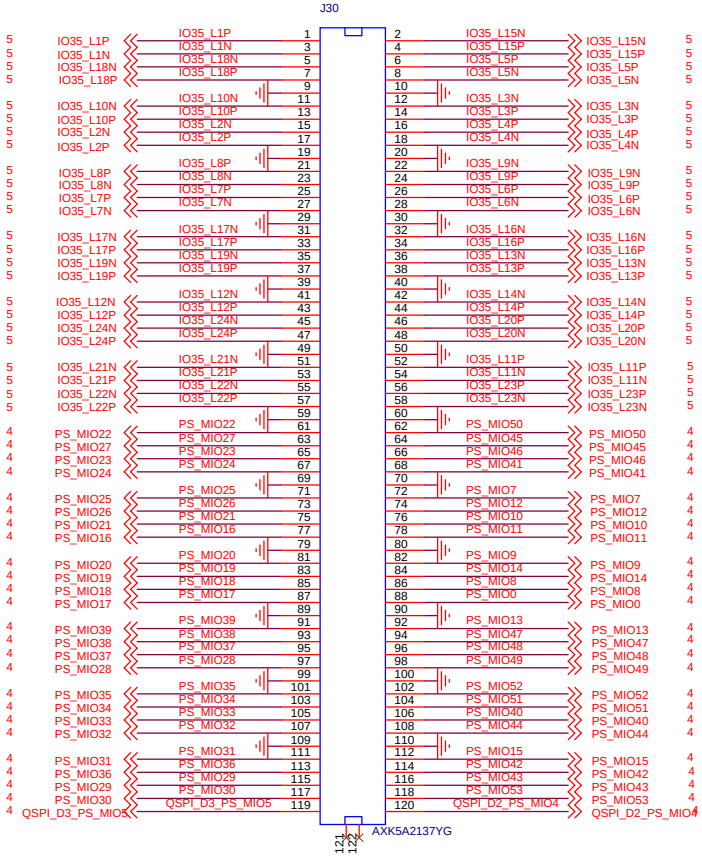
Table 5-12: DDR Routing Topology

DDR POWER For VTT/VREF

Title	Zynq-7000 DDR		
Size	Document Number: AC7Z010/AC7Z020 核心板原理图		Rev: 1.0
Date: Monday, December 12, 2022	Sheet 8 of 10		

VCCIO34，VCCIO35的输入电源由外部提供，在上电顺序要求5V先供电，然后再供VCCIO34和VCCIO35





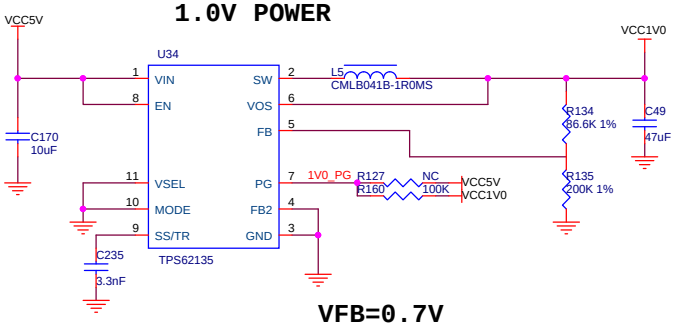
BANK13的IO管脚只有ACZ020的板子才有，ACZ010的核心板无法使用这些IO

PS_MIO0,MIO7~15为3.3V电平标准

PS_MIO16~53为1.8V电平标准

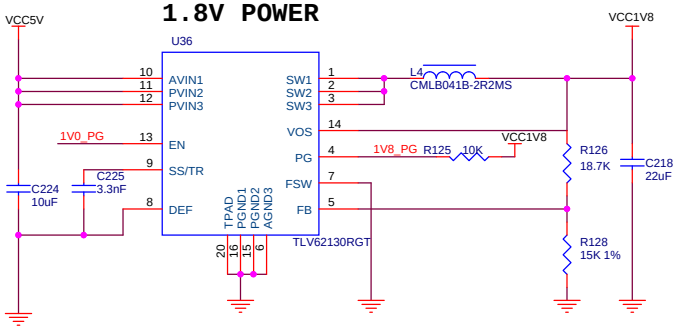
Title	Zynq-7000 LED & KEY & IO		
Size	Document Number		Rev
	AC7Z010/AC7Z020 核心板原理图		1.0
Date:	Wednesday, November 30, 2022	Sheet 9 of 10	

1.0V POWER

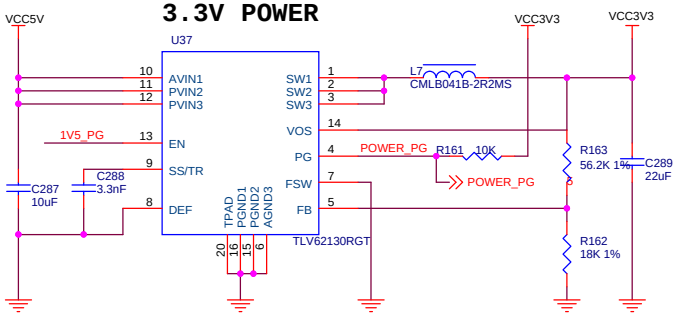


VFB=0.7V

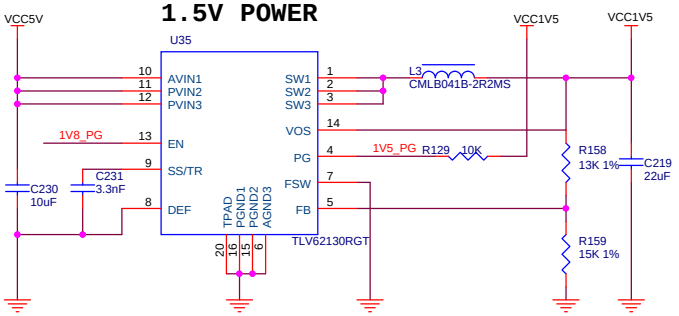
1.8V POWER



3.3V POWER



1.5V POWER



Power On Sequnce:

1.0V -> 1.8V -> 1.5 V -> 3.3V -> VCCIO

ALINX		
Title	Power	
Size	Document Number: AC7Z010/AC7Z020 核心板原理图	Rev: 1.0
Date: Wednesday, November 30, 2022	Sheet 10 of 10	